Certifying lossy decoding on the model's known benchmarks: same-scores gates for a 125B mixture-of-experts model on one 64 GB Mac

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Abstract

Lossy decoding is what makes a 125B mixture-of-experts model fast on one 64 GB Mac ([MAIN]), and it changes the output. We ask a narrow question: does a lossy configuration score the same as its exact reference on the model's known benchmarks?

- **The gate:** pre-registered and paired, one benchmark at a time.
  - The paired accuracy delta must be above −5 points, and the exact two-sided McNemar test must not reject (p >= 0.05).
  - It runs in two prompt orders, against the exact one-token greedy reference of the same memory mode.
- **Exactness** is audited separately: every token, and the logits row behind every kept token, must be bitwise equal to the reference's own one-token decode.
- **What passed, 20 October to 9 November 2026 [K]:**
  - L + COH, the 45 GB configuration we publish, and the chat configuration (M1-B + COH) passed the model card's IFBench, GPQA Diamond and LiveCodeBench v6 in both orders.
  - Its certified base ("v1' + wired"; L adds to it a bundle of kernel changes that is bitwise in the exact mode) passed all 10 of its pre-registered lines: HumanEval+, MBPP, math and IFEval in both orders, a pooled line and a long-code line. A certificate covers only the configuration it tested, so it does not carry to L + COH.
  - Exact decode paths that differ only in numerics (REF2, REF2b, NAX6) passed the same gate against each other.
  - Four more lossy configurations passed on the benchmarks they were gated on: 8-bit mixers; a short stack gate; a 32 GB point on HumanEval+ only; the chat configuration on IFEval (before its card gate).
- **What failed [K]:**
  - six quality-defined modes;
  - a code profile on the −5-point boundary;
  - every 12 GB candidate we gated on task scores.
- **Where we lose:**
  - The gate is coarse. Exact paths that differ only in numerics moved one benchmark by 8 points on 50 items [K]. An arm with no true loss fails one order of one benchmark with a probability we price at 5-15 % [P].
  - Most of our gates ran on subsets of public benchmarks and on home-made lines. The model card's own benchmarks (IFBench, GPQA Diamond, LiveCodeBench v6) were adopted as the gate on 8 November, on 381 keys: IFBench 150 of 300, GPQA Diamond 100 of 198 and one 131-problem LiveCodeBench window. L + COH has passed them, and so has the chat configuration (M1-B + COH; 6 of 6 lines); the 32 GB point (zs32) is not gated (no card gate ran).
  - We list our mistakes, among them a reference switched after we had seen a result, and a chat path we called exact without having measured it.

Marks: [K] measured, [P] priced, [G] judgement. Each number carries a source key (Appendix A). Every number below was re-read from its source on 8 or 9 Nov 2026. None is quoted from a summary.

1. Introduction

The setting. The engine of [MAIN] runs a 125B mixture-of-experts model on one 64 GB Mac.

- It has an exact mode and lossy modes. In the exact mode, every token is the reference's one-token greedy choice.
- The lossy modes are faster because they skip, approximate or reuse work: drafts accepted on a weaker check, expert computation at lower precision, and quantization tiers.
- A lossy mode is only useful if it keeps the model's abilities. "Keeps its abilities" is easy to claim and hard to check.

The narrow question. We do not ask whether a lossy mode produces the same tokens; it does not. We ask whether it scores the same as its exact reference on benchmarks the model is known by:

- paired, prompt by prompt;
- with a fixed, pre-registered threshold;
- in more than one prompt order, because a lossy mode's output can depend on what ran before it (section 4.2).

Contributions:

1. A gate protocol: the rule, two histories, the own-mode exact reference, and pre-registration with disclosed amendments (section 4).
2. An exactness audit that makes "exact" a measured property, not a label (section 3).
3. About 30 pre-registered verdicts on one model and one machine, passes and failures alike (section 5).
4. A catalogue of our own mistakes and the fix each led to (section 6).
5. A move, mid-project, from home-made lines to the model card's benchmarks, with the reasons and the cost (sections 4.5 and 8).

What we do not claim.

- A pass is "same scores on the named benchmarks, at the stated N, in both orders". It does not claim unchanged quality, and it does not mean there is no loss.
- A gate at 100-164 items cannot see a loss of 2-3 points (section 4.6).
- Exact means exact to a named reference (45 GB: NAX6-N), not to the original 4-bit weights ([MAIN] section 2).

2. Setting

- **Model and machine:**
  - Qwen3.8-Flash-Next: 125B parameters, mixture of experts, 48 layers, 512 routed experts per layer, 10 per token.
  - One MacBook Pro (M5 Pro, 18-core CPU, 20-core GPU, 64 GB, 2 TB SSD), with unified memory and the internal SSD.
  - The store holds about 3.2 bits per expert weight: ranks 0-1 at 4 bits, ranks 2-9 at 3 bits ([MAIN] section 2).
- **Memory lines:** a run is on the N GB line when its lifetime peak footprint is at most N GB.
  - GB = `10^9` bytes of the lifetime peak (ri_lifetime_max_phys_footprint), so the 45 GB line is 41.9 GiB [K, ES:evalsuite/decode.py footprint_peak_gb]; a 7.45 GB expert tier held as purgeable memory is not counted in it ([MAIN] section 2).
  - The line is the engine's own footprint, not an OS limit.
- **The exact reference chain [K]:**
  - The stock reference: the original 4-bit weights, decoded 20 October.
  - REF2, a batch-invariant exact path ([KERNELS]).
  - REF2b, which adds key buckets.
  - REF2-T, which adds the store's 3-bit tiers. The tiers are lossy against REF2b; REF2-T passed the gate and became the reference.
  - NAX6, the matrix-unit expert kernel, and NAX6-N, the same on the converted store. NAX6-N is the 45 GB exact reference since 5 November.
  - Each step passed the same gate as a lossy arm would (Table 1a). Each exact path was audited bitwise (Table 1b).
- **Lossy levers.** These are described where they are built: [MAIN] section 5 for the 45 / 32 GB levers, a forthcoming companion paper ([SMALL-RAM], by 11 November 2026) for the 24 / 12 GB loops. Here they are arms. A configuration is a frozen code snapshot plus an environment file, named in each result's source.

- **Decoding:** greedy, non-thinking chat template, one prompt at a time, with fixed caps [K, ES:evalsuite/data.py; tools/mbpp_build.py; tools/card_items.py]:
  - MATH-500 1,536 and GSM8K 768;
  - IFEval 1,024;
  - HumanEval+ 768;
  - MBPP 1,024;
  - IFBench 1,536;
  - GPQA Diamond 2,048;
  - LiveCodeBench v6 2,048.

3. Exactness is audited, not assumed (Table 1b)

The audit. "Exact (to R)" means that the output tokens, and the logits row behind every kept token, are bit-identical to one-token greedy decoding under R ([MAIN] section 2). We test exactly that.

- A fixed set of 36 full-length prompts: 12 code, 12 IFEval, 12 math.
- Each is decoded by the arm's own one-token AR and by its live 16-row drafting loop.
- Every kept logits row is compared bitwise. A prompt whose AR leaves the stored reference rows is VOID.

The first audit failed, and that was useful.

- R-56 tested REF2's live loop: 28 of 36 prompts were exact, and math only 4 of 12 [K, EV:RESULT_r56].
- Below the AR's 512-key point, 2,153 of 2,199 compared rows were bitwise equal. Past it, the attention bits followed the key length, and tokens flipped only at ties [K, EV:RESULT_r56].
- An earlier claim that REF2 was "bit-exact inside the live loop" had compared its equal prompts over their first 48-97 positions only.
- The fix (key buckets: REF2b) then passed 36 of 36, with 16,305 of 16,305 kept rows bitwise [K, EV:RESULT_r56b].
- Every later exact path was audited the same way before it was used as a reference: REF2-T; REF2-T with out-of-order experts and a fused combine; NAX6-N (Table 1b).
- A speed change that must not change outputs gets the same kind of audit against its own control. A PASS lets it inherit its configuration's gate verdicts without a gate of its own. Examples: two changes in [RELEASE]'s 45 GB chat configuration, the prefill staging PF-D v4 and the schedule change SB1 (Table 1b).

Line checks. A reference decoded at a smaller expert cache (C 160 instead of 192) is used only after a line check. 12 IFEval rows at that C must be token-identical to the rows at C 192; at C 160 they were (12 / 12) [K, ES:results/card/linecheck_C160.txt].

Three words kept apart.

- *Exact* (to R): bitwise, as above.
- *Verified*: every committed token is the full model's greedy choice at its computed width, but not bit-identical to one-token decoding. This is the 12 GB loop (described in the forthcoming [SMALL-RAM]).
- *Deterministic*: the same output on a rerun.
- We conflated them once (section 6, item 9).

4. The same-scores gate

4.1 The rule

"Decision 4" has been used since 28 October [K, EV:PREREG_ref2_2026-10-28].

- In each named benchmark, the paired delta (arm − reference) must be above −5 points, AND the exact two-sided McNemar p must be at least 0.05.
- A benchmark fails if either part fails. The configuration fails if any benchmark fails.
- A gain with $p < 0.05$ is reported as "DIFFERS (HIGHER)" and counts as a pass.
- The floor (the same weights with only the summation order changed) is reported beside each verdict as the noise yardstick.

What it replaced. The suite's first rule set (19-20 October) had 5-17 lines [K, ES:docs/DESIGN.md]:

- per-set deltas, answer agreement, length and hesitation ratios, and "no rise in truncations or loops".
- The truncation / loop line failed the stock reference against a re-decode of itself, although 200 of its 210 rows were token-identical [K, EV:RESULT_ref2]. It became information only.

4.2 Two prompt orders

Why. A lossy mode's output can depend on history.

- **Cache state:** which experts earlier prompts left resident. Two runs of one frugal 12 GB configuration differed "by up to" 4 points per set [K, EV:RESULT_smallram].
- **The drafts:** on long agent turns, two orders gave token-identical rows on 3 of 17 turns [K, EV:RESULT_agentfast].
- Since 3-6 November, every lossy gate runs the keys in two orders (sorted forward, then reversed) and must pass in both [K, EV:PREREG_TEMPLATE_cachedep; EV:PREREG_stack_2026-11-06].
- **One order is allowed only after a history-independence test.** The 12 GB IFBench arm's harness decoded 13 prompts in another order and matched bit for bit, with SSD bytes moving −1.1 % to +12.5 % between orders [K, F12H].
- The template also asked for a repeat check: the same history must reproduce the same tokens. It appears in no later pre-registration [K]. This is a gap (section 8).

4.3 The reference of the arm's own mode

The rule. A lossy arm is compared with the exact reference of its own memory mode: NAX6-N at 45 GB, since 5 November. It is not compared with an older or a different exact path. Exact paths themselves differ on task scores by more than the gate's resolution (Table 1a).

The one time this rule moved after a result.

- On 7 November the FAST code profile ("STACK") failed against REF2-T on HumanEval+ by the smallest margin: −5.00 points in both orders, p 0.25 [K, EV:RESULT_stack].
- The next gate's reference was then changed to NAX6-N, the stack's own exact base. The amendment says so: "Made after seeing a result" [K, EV:PREREG_s4].
- The rule itself had been in force since 5 November. The disclosure stays. S4, C30 and CERT's HumanEval+ forward order all fail against REF2-T as information (Table 2).
- The 12 GB gates still compare with REF2-T (section 8).

4.4 Pre-registration

- Every gate has a PREREG document committed before its first row. It names the arm, the reference, the sets and N, the orders, the rule and the stop conditions.
- Amendments are dated and committed before the rows they affect. Scheduling-only amendments say so. Rule changes say when they were made relative to the data.
- **Exceptions disclosed in our records [K]:**
  - one amendment made after 80 rows, with only row counts seen [EV:PREREG_pmin12];
  - the reference change above, made after a result [EV:PREREG_s4];
  - one arm reported as "one of three tries" on the same 60 prompts [EV:PREREG_night_gates].

4.5 Benchmarks: from subsets and home-made lines to the model card's

Until 8 November.

- The gates ran on fixed subsets of public benchmarks:
  - math, 50 problems from MATH-500 levels 3-5 and GSM8K;
  - IFEval, 100 prompts, strict, prompt level;
  - HumanEval+, 60 problems, pass@1;
  - later MBPP 100, HumanEval+ 164 and math 100 [K, EV:RESULT_cert].
- Beside them ran home-made lines that sometimes decided verdicts:
  - blunders per 10,000 tokens on long agent and code turns;
  - KLD drift;
  - a multilingual blunder line.

The 8 November rule. Lossy configurations are judged against exact on the model's known benchmarks. Home-made lines (blunders, KLD, multilingual drift) are information only [K, EV:PREREG_chat_gate].

The card gates [K, EV:PREREG_card_gates]:

- **Sets:** the model card's three public benchmarks.
  - IFBench (300 prompts; the official grader vendored; prompt-level loose, the paper's headline metric).
  - GPQA Diamond (198; generative, with the simple-evals template and answer pattern).
  - LiveCodeBench v6: the 131 problems of the 2025-02 to 2025-05 window that the card's family names. LiveCodeBench's own run_test grader is vendored and runs on public and private tests.
- **Subsets:** the gate keys are seeded subsets, IFBench 150 and GPQA Diamond 100 (seed 1108), plus the whole LCB window: 381 keys. They were sized so that one arm in two orders fits one evening window: about 450K tokens and 3.6 h [P, EV:PREREG_card_gates]. The first arm, L + COH, decoded 867,659 tokens in 5.8 h of decoding (forward 164 min, reverse 176 min, re-runs 6 min; process starts included) [K, EV:RESULT_card_lcoh_2026-11-09]. Its rate, about 42 tok/s with the starts, was within the priced 38-45 tok/s. Its rows were about twice as long as priced (1,139 tokens per row against about 590), so an arm needs a night, not an evening.
- **The extension rule:** an arm that fails one order only on the −5-point delta, with p >= 0.05, is PENDING. That benchmark is extended to its full set in both orders before it counts as a fail. A McNemar $p < 0.05$ stays a fail at once.
- **What the card gates do not test:** the card's numbers (81.3, 91.9 and 91.7) are thinking mode with sampling. Ours are non-thinking greedy, on quantized weights, so they cannot be compared with the card at fine resolution (section 8).
- **The caps bind on two sets** [K, ES:results/ref2tnaxn.r6/decode.jsonl]. The exact reference stops at its cap on 54 of 100 GPQA Diamond rows and 88 of 131 LiveCodeBench rows (IFBench: 10 of 150). Arm and reference run under the same caps. So the gate asks whether a lossy arm keeps the reference's results within that budget, not what the model scores at full length. A lever that makes answers longer would lose rows to the cap, and that would count as a loss.

4.6 Power: what the gate can and cannot see

The noise floor, measured. Between exact paths that differ only in numerics [K, Table 1a]:

- math moved by up to 8 points on 50 items (REF2 vs the stock re-decode);
- IFEval moved by up to 5 points on 100 items (REF2c-T vs REF2-T, a FAIL on the boundary). The same pair passed on 220 fresh IFEval prompts (+1.36 points, p 0.51) [K, EV:RESULT_ifx].

Priced for the card gates [P, EV:PREREG_card_gates "Known limits"]. Take an arm with no true loss whose rows differ from the reference on 10-15 % of keys.

- The −5-point line fails it in one order with these chances:
  - ~5 % at 10 % discordance (N 100);
  - ~15 % at 15 % discordance (N 100);
  - ~7 % at 15 % discordance (N 150).
- Over 3 benchmarks × 2 orders, it fails somewhere with a real chance, ~25-50 %.
- The full sets would cut that to ~2-5 % per order, at about +236K tokens per arm.
- The extension rule (4.5) is our answer to that cost.

The boundary. Deltas of exactly −5.0 fail. Four of our fails sit on it (Table 3, item 6).

5. Results

Table 1a: exact paths, same-scores gate against each other

Core 210 = math 50, IFEval 100, HumanEval+ 60. One order. Cells give arm vs reference accuracy %, the paired delta in points, and McNemar p. All [K].

arm (kind)	vs	math	IFEval	HumanEval+	verdict	source
REF2 (batch-invariant exact path)	stock reference, re-decoded	90.0 vs 82.0, +8.0, p 0.12	87.0 vs 90.0, −3.0, p 0.38	91.7 vs 91.7, 0.0, p 1.00	PASS; token-identical 68/210 (floor 59/210)	EV:RESULT_ ref2_2026-10-28

arm (kind)	vs	math	IFEval	HumanEval+	verdict	source
REF2b (+ key buckets)	stock re-decode	88.0 vs 82.0, +6.0, p 0.25	87.0 vs 90.0, −3.0, p 0.38	91.7 vs 91.7, 0.0, p 1.00	PASS	EV:RESULT_ ref2b_2026-10-31
REF2-T (+ the store's 3-bit tiers; lossy vs REF2b, then the reference)	REF2b	88.0 vs 88.0, 0.0, p 1.00	90.0 vs 87.0, +3.0, p 0.38	95.0 vs 91.7, +3.3, p 0.50	PASS; token-identical 48/210 (floor 63/210)	EV:RESULT_ r58_2026-10-31
REF2c-T (another exact path)	REF2-T	86.0 vs 88.0, −2.0, p 1.00	85.0 vs 90.0, −5.0, p 0.0625: FAIL	93.3 vs 95.0, −1.7, p 1.00	FAIL on the boundary	EV:RESULT_ ref2ct_2026-11-02
REF2c-T, IFEval re-test on 220 fresh prompts	REF2-T		84.5 vs 83.2, +1.36, p 0.51 (corrected 11-04; verdict unchanged)		PASS	EV:RESULT_ifx_ 2026-11-03
NAX6 (matrix-unit expert kernel)	REF2-T	88.0 vs 88.0, +0.0, p 1.00	87.0 vs 90.0, −3.0, p 0.375	91.7 vs 95.0, −3.3, p 0.50	PASS; the 45 GB reference	EV:RESULT_ nax6_2026-11-05

Table 1b: exactness audits

Bitwise against the arm's own one-token AR, on the 36 full-length prompts. The last three rows compare against a named control instead.

audit	result [K]	verdict	source
R-56: REF2's 16-row live loop	28/36 prompts (math 4/12); bitwise below the 512-key point: 2,153 of 2,199 rows	FAIL	EV:RESULT_r56_2026-10-30
R-56b: REF2b (also with the sync-free gate)	36/36, 16,341 tokens, 16,305 / 16,305 kept rows	PASS	EV:RESULT_r56b_2026-10-31
R-56c: REF2-T's live loop	36/36, 15,525 tokens, 15,489 / 15,489 kept rows	PASS	EV:RESULT_r56c_2026-11-01
R-56d: REF2-T + out-of-order experts + fused combine	36/36; loop = own AR = R-56c's AR; 15,489 / 15,489 rows	PASS	EV:RESULT_r56d_2026-11-01
NAX6-N	36/36, 15,726 tokens, 15,690 / 15,690 kept rows; NAX6-N == NAX6	PASS	EV:RESULT_nax6_2026-11-05
card reference line check, C 160 vs C 192	12 / 12 IFEval rows token-identical	PASS	ES:results/card/linecheck_ C160.txt
PF-D v4 + read-ahead: chat prefill takes experts already in the pool; on by default in the 45 GB chat configuration (zs: M1-B + COH + SB1, C 256)	vs its own control with PF-D off, 12 IFEval prompts: tokens 12 / 12, logits hashes 721 / 721. With its check on (exact path E): 87,912 staged rows, 0 differing	PASS (bitwise to its control)	BM:data/pfd/pfd_report_ 0009.txt; BM:data/r57/pdz4a_ 256_runs.json, pdz0_256_ runs.json, pde4k_256_runs.json
SB1, a schedule change in the chat configuration (zs = M1-B + COH + SB1)	tokens: zs, and zs + PF-D v4 as shipped, equal M1-B + COH on 12 / 12 IFEval prompts (C 256, one history). Logits on the exact path E: E + SB1 vs E, 676 / 676 hashes equal (12 prompts). On the chat configuration the link is token-level only: those runs kept no logits hashes	PASS (tokens; logits on E)	BM:data/r57/z0_256_tokens.json, zs_256_tokens.json, pdz4a_256_ tokens.json; BM:data/r57/yeh_ 256_runs.json, yesh_256_ runs.json; BM:docs/WED_ TESTS.md (CHS5)

Table 2: lossy configurations

The 45 GB reference is REF2-T until 7 November and NAX6-N from then on (section 4.3). Deltas are paired points, and p is McNemar's. All [K].

45 / 32 GB lines

configuration	vs	sets (n) × orders	result	verdict	source
HC mixers at 8 bits (HC8)	REF2-T	core 210 × 1	math −4.0, p 0.50; IFEval −1.0, p 1.00; HumanEval+ −3.3, p 0.50	PASS	EV:RESULT_ hc8_2026-11-02
RELAX ship mode (code mode B + chat mode E)	REF2-T	core 210 × 1	math +2.0; IFEval −3.0, p 0.25; HumanEval+ −1.7. Decision 4 PASS; agent-blunder and drift bars failed	BETWEEN	EV:RESULT_ relax_2026-11-05
RELAX A_E / C_E / B_F / B_G / T3ALL	REF2-T	core 210 × 1	math −6.0, p 0.25 / HumanEval+ −5.0, p 0.25 / IFEval −5.0, p 0.12 / IFEval −6.0, p 0.07 / HumanEval+ −6.7, p 0.12	KILL (all five)	EV:RESULT_ relax_2026-11-05
ship8 (B + E8; HC8 process-global)	REF2-T	core 210 × 1	IFEval 84.0 vs 90.0, −6.0, p 0.03	KILL	EV:RESULT_ relax_drift_ 2026-11-06
STACK (FAST code profile, C 224)	REF2-T	math 50 + HumanEval+ 60 × 2	HumanEval+ 90.0 vs 95.0, −5.00, p 0.25 in both orders	FAIL (kept on record)	EV:RESULT_ stack_2026-11-07
STACK without CD16, MBPP	NAX6-N	MBPP 100 × 2	76.0 vs 79.0, −3.00, p 0.4531 in both orders	PASS	EV:RESULT_ stack_2026-11-07
S4 v1	NAX6-N	math 50 + HumanEval+ 60 × 2	math 86.0 vs 88.0, −2.00, p 1.0; HumanEval+ 88.3 vs 91.7, −3.33, p 0.50 (both orders)	PASS; vs REF2-T HumanEval+ FAIL (information)	EV:RESULT_s4_ 2026-11-07
C30: L at C 160, before COH (the ~32 GB lossy context point)	NAX6-N	HumanEval+ 60 × 2	forward −3.33, p 0.625; reverse +0.00, p 1.0	PASS (HumanEval+ only); vs REF2-T forward −6.67: FAIL (information)	EV:RESULT_ c30_2026-11-07
CERT: "v1' + wired", the certified 45 GB configuration (C 256). L adds the exact-safe bundle (BM:data/ stack/s5p_env.txt; bitwise in the exact mode, [MAIN] 4.4); the published L + COH has its own gate (below)	NAX6-N	HumanEval+ 164; math, MBPP, IFEval 100 each; × 2	HumanEval+ −2.44, p 0.125 / +0.00, p 1.0. Math +0.00, p 1.0 / −2.00, p 0.625. MBPP −3.00, p 0.375 / +2.00, p 0.727. IFEval −1.00, p 1.0 / −2.00, p 0.754. Pooled over HumanEval+, math and MBPP in both orders (728 pairs; IFEval not pooled) −0.96, p 0.265. Long code (41 rows, 55,966 tokens): 0 blunders	PASS, 10 of 10 lines. Chat NOT exact: 1 of 12 rows identical	EV:RESULT_ cert_2026-11-08
chat configuration M1-B + COH	NAX6-N	IFEval 100 × 2	forward −1.00, p 1.0; reverse −4.00, p 0.388	PASS (IFEval)	EV:RESULT_ chat_m1b_ 2026-11-08
plain FAST on long agent turns	NAX6-N's verify path (a home-made line)	17 turns × 2	85 blunders over 34 rows, 19,749 tokens; the line was 0	FAIL; agent turns run exact	EV:RESULT_ agentfast_ 2026-11-06

24 / 12 GB lines and the store

configuration	vs	sets × orders	result	verdict	source
the store's tier plan (ranks 2-9 at 3 bits)	the original 4-bit weights	math 50, IFEval 100, HumanEval+ 60, RULER 24 × 1	+0.0, p 1.00 / −4.0, p 0.29 / +0.0, p 1.00 / +4.2, p 1.00; loops 4 vs 2 (limit 3)	16 / 17 lines; adopted 20 Oct	EV:RESULT_ plan4433333333; TR:evalsuite.md
T0: 12 GB two-pass top-2	stock reference	core 210 × 1	math −2.0, p 1.00; IFEval −6.0, p 0.07; HumanEval+ −11.7, p 0.02	KILL	EV:RESULT_t0_ 2026-11-05

configuration	vs	sets × orders	result	verdict	source
12 GB FAST family, θ 0.8 / 0.9 / 0.95	REF2-T	HumanEval+ 60 × 2	−11.67, p 0.0156 / −8.33, p 0.0625 / −5.00, p 0.25	FAIL (all three)	EV:RESULT_ t80ck1u_2026-11-07
PMIN12: a cut of the MTP chain	its control	239 rows × 1	identical scores; peak 11.99 GB vs the 11.95 GB line	FAIL (memory only)	EV:RESULT_ pmin12_2026-11-01
fold-fast	its exact twin	239 rows × 1	answer agreement 210/210; peaks up to 15.60 GB	not PASS (memory)	EV:RESULT_r41c_ 2026-10-28
24 GB fast mode b16	its natural-routing base / the stock reference	core 210 + MGSM 80 × 1	IFEval −4.0 vs base; −5.0 vs stock	5/5 vs base; 4/5 vs stock (FAIL)	EV:RESULT_ fastmode_b16
small-RAM family (older rule set; two runs pooled)	the stock reference	core 210 × 1-2	e.g. B12LL-b16 IFEval −7.0, p 0.02; 24 GB D1 pooled IFEval −6.0	mixed; "READ WITH NOISE"	EV:RESULT_ smallram_2026-10-22

The card gates (the model card's benchmarks)

item	state	source
NAX6-N reference on the 381 gate keys	complete, 381 / 381 rows [K] (9 Nov). Rows at the cap: IFBench 10 / 150, GPQA Diamond 54 / 100, LCB v6 88 / 131 [K]	ES:results/ref2tnaxn.r6; EV:PREREG_ card_gates
L + COH (the published 45 GB configuration; QOM on LCB, exact expert reads on IFBench and GPQA Diamond)	PASS, 6 of 6 lines [K] (9 Nov): lossy, same scores on IFBench, GPQA Diamond and LCB v6. IFBench 40.7 vs 44.0 %, −3.33, p 0.302 in both orders. GPQA Diamond 47.0 vs 44.0 %, +3.00, p 0.508 in both. LCB v6 30.5 vs 32.1 %, −1.53, p 0.688 / 29.0 %, −3.05, p 0.289. On IFBench and GPQA the two orders gave the same tokens on every row	EV:RESULT_card_lcoh_2026-11-09
M1-B + COH (the chat configuration; the shipped 45 GB chat configuration adds SB1 and PF-D v4, both audited in Table 1b)	PASS, 6 of 6 lines [K] (9 Nov): lossy, same scores on IFBench, GPQA Diamond and LCB v6. Forward / reverse against 44.0, 44.0 and 32.1 %: IFBench 44.0 / 41.3 % (p 1.0 / 0.50), GPQA Diamond 42.0 / 42.0 % (p 0.75 / 0.79), LCB v6 29.0 / 32.1 % (p 0.39 / 1.0)	EV:RESULT_card_m1b_2026-11-09; EV:PREREG_card_gates
zs32: M1-B + COH + SB1 at C 160 (32 GB)	not gated: no card gate ran (amendment 11)	EV:PREREG_card_gates amendments 8 and 11

6. Pitfalls and our mistakes (Table 3)

All items come from our own records [K]. Each led to a rule or a tool that is still in use.

#	what happened	consequence / fix	source
1	The stock reference did not reproduce on later code: rounding changed on 10 of 210 rows between two engine versions	the re-decode became the baseline; the cause was not looked into	EV:RESULT_ref2_2026-10-28
2	The truncation / loop line failed the stock reference against a re-decode of itself	demoted to information	EV:RESULT_ref2_2026-10-28
3	The vendored IFEval checker called its language detector unseeded, so a re-score could move IFEval by one item	seeded; all tables re-scored, no verdict moved	EV:RESULT_smallram_2026-10-22
4	IFEval's letter-frequency check with a non-letter ("!") scored a randomly drawn letter	one re-test corrected (84.1 → 84.5, p 0.75 → 0.51); verdict unchanged	EV:RESULT_ifx_2026-11-03
5	"Bit-exact inside the live loop" had been checked on the first 48-97 positions only; it fails past 512 keys	key buckets, and bitwise audits of every kept row (section 3)	EV:RESULT_r56_2026-10-30

#	what happened	consequence / fix	source
6	Deltas of exactly −5.0 fail the line (REF2c-T IFEval, STACK HumanEval+, the θ 0.95 arm, RELAX C_E and B_F). REF2c-T then passed on 220 fresh prompts	a re-test stage; the card gates' PENDING / extension rule	EV:RESULT_ref2ct, RESULT_ifx, PREREG_card_gates
7	The reference was changed after a result (STACK → S4's NAX6-N)	disclosed; both comparisons reported (section 4.3)	EV:PREREG_s4_2026-11-07
8	"Chat stays exact" in the stack was assumed, not measured	CERT measured it: not exact (1 of 12 rows); the wording became "lossy, same scores"	EV:PREREG_cert, RESULT_cert
9	The 12 GB "exact" mode is verified, not exact, and "×1.12 over exact" was measured against a FAST arm (40.8 tok/s)	headline withdrawn; a 32-token setup check	EV:PREREG_t80ck1u_2026-11-07
10	The HC8 switch was process-global as built, so chat ran HC8 although only code was gated	the ship8 KILL	EV:RESULT_relax_drift_2026-11-06
11	Live code changed mid-run: a commit landed between the pin check and the import	the chunk was voided; frozen snapshots with per-process module manifests	EV:PREREG_agentfast_2026-11-06
12	A dry run took the real path: an apostrophe inside a shell ${...:?} message swallowed lines	2 rows voided; rule: no apostrophe inside ${} text	EV:PREREG_chat_gate_2026-11-08
13	Home-made lines (blunders, KLD, multilingual drift) decided verdicts until 8 November	information only; gates use known benchmarks	EV:PREREG_chat_gate, PREREG_card_gates
14	A cheap screen (the decision map) predicted PASS; the decode failed (12 GB θ family)	screens rank, gates decide	EV:RESULT_t80ck1u_2026-11-07
15	A cutoff bounded chunk starts, not ends, so a chunk ran past its window	cutoffs now bound chunk ends	EV:RESULT_relax_2026-11-05
16	On the shared machine, a reference runner read another job's "ended, incomplete" signal as "done", and a waiting decode client started 30 s into a heat pause	only a finished signal releases it; a guard stops clients during a pause (no row was affected)	EV:PREREG_card_gates amendments 4, 6
17	The arm screen stops a row mid-decode when its process's 600 s budget runs out, and flags it. The row writer did not read the flag and kept 10 such rows in the first card arm's forward order, ending them as "length" below the cap. The engagement check caught a later one (a 1-token LCB row with no QOM use) and voided the arm	cut rows are left out and their keys re-run; the arm resumed before any verdict. An audit of 40 earlier runs found no verdict that used a cut row	EV:PREREG_card_gates amendment 9

7. Related work

Nothing here claims priority. Each arXiv id below resolves to the named title and first author (arXiv API, 9 Nov); both repositories are MIT-licensed; the three journal papers match their Crossref records; the blog post and llama.cpp's KL-divergence mode were opened.

- **Evaluation harnesses:**
  - EleutherAI's lm-evaluation-harness (github.com/EleutherAI/lm-evaluation-harness; Biderman et al., arXiv 2405.14782);
  - HELM (Liang et al., arXiv 2211.09110).
- **The benchmarks:**
  - HumanEval (Chen et al., arXiv 2107.03374); EvalPlus / HumanEval+ (Liu et al., arXiv 2305.01210); MBPP (Austin et al., arXiv 2108.07732);
  - MATH (Hendrycks et al., arXiv 2103.03874), with the MATH-500 split of Lightman et al. (arXiv 2305.20050); GSM8K (Cobbe et al., arXiv 2110.14168);
  - IFEval (Zhou et al., arXiv 2311.07911); IFBench (Pyatkin et al., arXiv 2507.02833); GPQA (Rein et al., arXiv 2311.12022); LiveCodeBench (Jain et al., arXiv 2403.07974);
  - simple-evals (OpenAI; github.com/openai/simple-evals) for the GPQA template.
- **Quantization evaluation beyond average accuracy:**
  - "Accuracy is Not All You Need" (Dutta et al., arXiv 2407.09141), on flips;
  - accuracy-recovery studies of quantized LLMs, for example Kurtic et al. (arXiv 2411.02355);

  - KLD-based checks in local-inference tools (llama.cpp's perplexity tool has a KL-divergence mode).
- **Paired tests:**
  - McNemar (1947), Psychometrika 12(2);
  - Dietterich (1998), Neural Computation 10(7), on comparing classifiers;
  - non-inferiority and TOST (Schuirmann 1987, Journal of Pharmacokinetics and Biopharmaceutics 15(6)).
- **Exactness of fast decoding:**
  - speculative decoding (Leviathan et al., arXiv 2211.17192; Chen et al., arXiv 2302.01318), whose exactness is distributional;
  - batch invariance and nondeterminism in LLM inference (Thinking Machines Lab, "Defeating Nondeterminism in LLM Inference", Sep 2025; [KERNELS] has the kernel side).

8. What we left on the table

- **Power** [P, EV:PREREG_card_gates]. Full card sets (IFBench 300, GPQA Diamond 198, LiveCodeBench v6 175) cut an arm with no true loss's chance to fail somewhere from about 25-50 % to about 2-5 % per order. They cost about +236K tokens per arm (about +1.7 h) and +118K on the reference (about +2.5 h). These prices predate the first arm, which decoded about twice the tokens priced for it (section 4.5), so they are likely low.
- **Longer caps** [P, ES:results/ref2tnaxn.r6/decode.jsonl]. The exact reference stops at the 2,048-token cap on 142 GPQA Diamond and LCB rows (section 4.5). Doubling those caps to 4,096 would add at most 290,816 tokens to the reference: 5.7-6.5 h at its measured 14.2 tok/s without heat pauses and 12.5 tok/s with them, plus each arm's share.
- **The card's own setting** [P, EV:PREREG_card_gates "The card anchor"]. The card's numbers are thinking mode with sampling.
  - Pricing exact in that mode with assumed lengths gives about 4.6M tokens, about 90 h of the exact reference at about 14 tok/s.
  - We did not run it. Our gates are non-thinking greedy.
- **The repeat check** of the two-history rule: the same history must reproduce the same tokens. It is in the template but in no later pre-registration [K, EV:PREREG_TEMPLATE_cachedep].
- **The 12 GB line's reference:** its gates compare with REF2-T, not with an exact reference of its own mode (its loop is verified, not exact). This is open.
- **The 12 GB IFBench arm:** its harness passed the history test (one order allowed) [K, F12H]. It runs after the release.
- **The multilingual question:** MGSM's place under the 8 November rule (a known benchmark, but the card does not report it) is open, and that gate is on hold.

Appendix A: sources

Keys follow [MAIN] Appendix C. Paths are relative to the root of the public snapshot, github.com/kamenlevi/350-tps; commit hashes are those of each folder's own repository, and the snapshot holds the files as last committed. The keys:

- EV: = EvalSuite/docs/ (RESULT_, *PREREG_*);
- ES: = EvalSuite/ (code, results/);
- F12H = the 12 GB track's history test, Fit12/data/scr/chw18;
- BM: = BigMoE/, the engine;
- TR: = FableGod/TRIED/, the attempts logs.

Per-row data: each RESULT names its results//decode.jsonl and report files. In the public snapshot, the answer text of every GPQA Diamond row in the decode.jsonl files under ES:results/ reads "(withheld: GPQA Diamond)": the answers often quote the questions, and the benchmark is kept out of training data. Token ids, timings and verdicts are unchanged.

- EV:RESULT_ref2_2026-10-28, RESULT_ref2b_2026-10-31, RESULT_r58_2026-10-31, RESULT_ref2ct_2026-11-02, RESULT_ifx_2026-11-03 (with its 11-04 correction), RESULT_nax6_2026-11-05.
- EV:RESULT_r56_2026-10-30, RESULT_r56b_2026-10-31, RESULT_r56c_2026-11-01, RESULT_r56d_2026-11-01.
- EV:RESULT_hc8_2026-11-02, RESULT_relax_2026-11-05, RESULT_relax_drift_2026-11-06, RESULT_agentfast_2026-11-06, RESULT_stack_2026-11-07, RESULT_s4_2026-11-07, RESULT_

c30_2026-11-07, RESULT_cert_2026-11-08, RESULT_chat_m1b_2026-11-08, RESULT_card_lcoh_2026-11-09, RESULT_card_m1b_2026-11-09.

- EV:RESULT_plan4433333333, RESULT_t0_2026-11-05, RESULT_t80ck1u_2026-11-07, RESULT_pmin12_2026-11-01, RESULT_r41c_2026-10-28, RESULT_fastmode_b16, RESULT_smallram_2026-10-22.
- EV:PREREG_ref2_2026-10-28 (decision 4), PREREG_TEMPLATE_cachedep, PREREG_stack_2026-11-06, PREREG_s4_2026-11-07, PREREG_cert_2026-11-08, PREREG_chat_gate_2026-11-08, PREREG_card_gates_2026-11-08 (amendments 1-11), PREREG_pmin12_2026-11-01, PREREG_night_gates_2026-11-07, PREREG_t80ck1u_2026-11-07, PREREG_agentfast_2026-11-06.
- ES:docs/DESIGN.md (the first rule set); ES:evalsuite/decode.py (footprint_peak_gb); ES:evalsuite/ifbench and ES:evalsuite/lcb (the vendored graders, with their licenses); ES:tools/card_items.py, tools/card_keys.json (the card sets and keys); ES:results/card/ (line check).
- F12H (chw18_eq.txt, chw18_hist.txt): the 12 GB history test.
- BM:data/pfd/pfd_report_0009.txt (lines pdz4a and pde4k) and BM:data/r57/pdz4a_256_tokens.json, pdz4a_256_runs.json, pdz0_256_tokens.json, pdz0_256_runs.json, pde4k_256_runs.json: the PF-D v4 audit. Re-counted from the runs files on 11-09: tokens 12 / 12, logits hashes 721 / 721, and summed check counters 87,912 rows with 0 differing.
- BM:data/r57/z0_256_tokens.json (M1-B + COH), zs_256_tokens.json (+ SB1), yeh_256_runs.json (E) and yesh_256_runs.json (E + SB1), with BM:docs/WED_TESTS.md, section CHS5: the SB1 audit. Re-counted on 11-09: zs, zs + PF-D v4, z0's repeat and PF-D's control all equal z0 on 12 / 12; E vs E + SB1, 676 / 676 logits hashes.
- TR:evalsuite.md: the attempts log (the store plan's adoption line).

Use of AI tools

The code, the experiments and the text of this paper were produced with substantial help from AI coding agents. The author directed the work and is responsible for its content. The numbers were re-derived from the raw run records named beside them, and the references were opened by hand or by script.

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Companion papers:

- [KERNELS] Kamen Levi. Batch-invariant kernels on Apple's M5 GPU and its matrix units. 2026. A companion paper of this set.
- [MAIN] Kamen Levi. What It Takes to Run a 125B LLM Fast on a 64 GB Mac, and How to Go Faster. 2026. A companion paper of this set.
- [RELEASE] Kamen Levi. Running a 125B mixture-of-experts model on a 64 GB Mac: the engine package. 2026. A companion paper of this set.
- [SMALL-RAM] Kamen Levi. A 125B mixture-of-experts model in 32, 24 and 12 GB of memory on one Mac. Forthcoming (by 11 November 2026).